

Further Statistics 1

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Chapter 7: Probability Generating Functions

Why this chapter matters

A probability generating function (PGF) packages an entire discrete probability distribution into a single algebraic function. This might seem like an abstract trick, but it turns hard problems (finding the mean/variance of a complicated distribution, or the distribution of a sum of independent variables) into comparatively easy algebra or calculus. PGFs are one of the most elegant tools in Further Statistics 1, and they connect directly to the sum-of-independent-variables results you met informally in earlier chapters.

Concept: definition of a probability generating function

For a discrete random variable X taking non-negative integer values, the probability generating function is defined as:

$$G_X(t) = E(t^X) = \sum P(X = x) \cdot t^x, \text{ summed over all possible values of } x.$$

The coefficient of t^x in the expansion of $G_X(t)$ is exactly $P(X = x)$ — this is why it's called a *generating* function, since expanding it “generates” the whole probability distribution.

What a PGF really “is”: think of $G_X(t)$ as a filing cabinet for the whole distribution — every probability $P(X = x)$ is tucked away as the coefficient of t^x , in one single expression, instead of being scattered across a table. You've already met this trick: the Binomial expansion $(q + pt)^n$ is literally a PGF, since expanding it by the binomial theorem produces exactly the terms ${}^nC_x p^x q^{n-x} t^x$ — read off the coefficient of t^x and you have $P(X = x)$ for $X \sim B(n, p)$ straight back again. The point of PGFs is that ordinary algebra and calculus applied to this one packaged object (multiplying two of them, or differentiating one) does the work that would otherwise mean manipulating an entire distribution term by term.

Key formula: PGFs of standard distributions

KEY FORMULA	
Distribution	PGF, $G_X(t)$
Binomial $B(n, p)$	$(q + pt)^n$, where $q = 1 - p$
Poisson $Po(\lambda)$	$e^{\lambda(t-1)}$
Geometric $Geo(p)$	$pt / (1 - qt)$, where $q = 1 - p$

You may be asked to derive any of these from the definition, so make sure you can reproduce each derivation, not just quote the result. The Geometric one is the quickest to derive and a good one to practise: $G_X(t) = \sum p(1-p)^{x-1} t^x$ for $x = 1, 2, 3, \dots = pt \sum (qt)^{x-1}$ (factoring out pt and writing $q = 1-p$) $= pt(1$

$+ qt + (qt)^2 + \dots = pt/(1-qt)$, using the infinite geometric series sum $1/(1-r)$ for $|r| < 1$ with $r = qt$ — the same summation formula from Core Pure/Year 1, just with an algebraic ratio instead of a numerical one.

Worked example 1: deriving a PGF from a distribution

EXAMPLE 7.1

Question. A discrete random variable X has $P(X = 0) = 1/4$, $P(X = 1) = 1/2$, $P(X = 2) = 1/4$. Find $G_X(t)$.

Step 1. Use the definition: $G_X(t) = \sum P(X=x)t^x = P(X=0)t^0 + P(X=1)t^1 + P(X=2)t^2$. **(M1)**

Step 2. Substitute: $G_X(t) = 1/4 + (1/2)t + (1/4)t^2$. **(A1)**

Answer: $G_X(t) = 1/4 + t/2 + t^2/4$

EXAM TIP

Check any PGF you derive using $G_X(1) = 1$ (since $\sum P(X=x) = 1$ when $t = 1$) — here $G_X(1) = 1/4 + 1/2 + 1/4 = 1$ ✓. This single check catches the majority of algebraic slips and should become an automatic habit, exactly like checking a probability distribution sums to 1 in Chapter 1.

Key formula: using the PGF to find mean and variance

The real power of a PGF is that its derivatives at $t = 1$ recover the mean and variance without ever summing the distribution directly:

KEY FORMULA

- $G_X'(1) = E(X)$
- $G_X''(1) = E[X(X-1)] = E(X^2) - E(X)$
- $\text{Var}(X) = G_X''(1) + G_X'(1) - [G_X'(1)]^2$

Why does differentiating and setting $t = 1$ recover $E(X)$? Differentiate the series $G_X(t) = \sum P(X=x)t^x$ term by term: each term $xP(X=x)t^{x-1}$ has picked up a factor of x from the power rule. So $G_X'(t) = \sum x \cdot P(X=x) \cdot t^{x-1}$, and setting $t = 1$ makes every t^{x-1} vanish to 1, leaving $G_X'(1) = \sum x \cdot P(X=x) = E(X)$ exactly, by the definition of expectation from Chapter 1. Differentiating a second time brings down another factor of $(x-1)$, which is why $G_X''(1)$ lands on $E[X(X-1)] = E(X^2) - E(X)$ rather than $E(X^2)$ directly — you then rearrange to get $E(X^2)$, and from there $\text{Var}(X)$ as usual.

Worked example 2: using a PGF to find $E(X)$ and $\text{Var}(X)$

EXAMPLE 7.2

Question. A discrete random variable X has PGF $G_X(t) = (1/8)(1+t)^3$. Find $E(X)$ and $\text{Var}(X)$.

Step 1. Differentiate: $G_X'(t) = (3/8)(1+t)^2$. **(M1A1)**

Step 2. Evaluate at $t = 1$: $G_X'(1) = (3/8)(2)^2 = (3/8)(4) = 3/2$, so $E(X) = 3/2$. **(A1)**

Step 3. Differentiate again: $G_X''(t) = (3/4)(1+t)$. **(M1A1)**

Step 4. Evaluate at $t = 1$: $G_X''(1) = (3/4)(2) = 3/2$. **(A1)**

Step 5. Apply the variance formula: $\text{Var}(X) = G_X''(1) + G_X'(1) - [G_X'(1)]^2 = 3/2 + 3/2 - (3/2)^2 = 3 - 9/4 = 3/4$. **(M1A1)**

Answer: $E(X) = 3/2$, $\text{Var}(X) = 3/4$ (in fact, $X \sim B(3, 1/2)$, and this matches the standard Binomial formulae $np = 3/2$ and $npq = 3 \times 0.5 \times 0.5 = 3/4$ — a good self-check.)

EXAM TIP

Where possible, recognise the underlying distribution first (here, $(1/8)(1+t)^3 = ((1+t)/2)^3$ matches $(q+pt)^n$ with $n=3$, $p=1/2$) so you can sanity-check the calculus answer against a known formula. This builds confidence that the PGF machinery is really just a more general route to results you can already verify, rather than mysterious new mathematics.

Concept: PGF of the sum of independent random variables

If X and Y are independent random variables with PGFs $G_X(t)$ and $G_Y(t)$, then the PGF of their sum is the *product* of the individual PGFs:

$$G_{X+Y}(t) = G_X(t) \times G_Y(t)$$

This is the single most important PGF result for exam purposes: it turns the (often difficult) problem of finding the distribution of a sum into simple algebraic multiplication.

Worked example 3: PGF of a sum of independent variables, identifying the resulting distribution

EXAMPLE 7.3

Question. $X \sim \text{Po}(2)$ and $Y \sim \text{Po}(5)$ are independent. Use probability generating functions to find the distribution of $X + Y$.

Step 1. Write down the PGFs: $G_X(t) = e^{2(t-1)}$, $G_Y(t) = e^{5(t-1)}$. **(B1)**

Step 2. Since X and Y are independent, $G_{X+Y}(t) = G_X(t) \times G_Y(t) = e^{2(t-1)} \times e^{5(t-1)} = e^{2(t-1)+5(t-1)} = e^{7(t-1)}$. **(M1A1)**

Step 3. Recognise this as the PGF of a Poisson distribution with parameter 7 (compare to the standard form $e^{\lambda(t-1)}$). **(M1)**

Answer: $X + Y \sim \text{Po}(7)$, confirming the additive property of independent Poisson variables met in Chapter 2, but now derived rigorously rather than just quoted.

EXAM TIP

This example closes the loop with Chapter 2: you accepted “sums of independent Poissons are Poisson” there on trust, and PGFs now let you actually prove it. Try the analogous proof for independent Binomial variables with the same p (also examinable) as an exercise — it reinforces both the PGF product rule and the uniqueness of PGFs (two variables with the same PGF must have the same distribution), which you should also know as a named fact.

Concept: uniqueness of the PGF

Two random variables have the same probability distribution if and only if they have the same PGF. This “uniqueness property” is what justifies the method in Example 3: once you’ve shown $G(X+Y)(t)$

matches the Poisson PGF form $e^{\lambda(t-1)}$ with $\lambda = 7$, you can conclude with certainty that $X + Y$ really does follow a $Po(7)$ distribution, not merely that it has a similar mean.

Chapter summary

Idea	Key point
Definition	$G_X(t) = E(t^X) = \sum P(X=x)t^x$
Standard PGFs	Binomial: $(q+pt)^n$; Poisson: $e^{\lambda(t-1)}$; Geometric: $pt/(1-qt)$
Check	$G_X(1) = 1$ always
Finding the mean	$E(X) = G_X'(1)$
Finding the variance	$\text{Var}(X) = G_X''(1) + G_X'(1) - [G_X'(1)]^2$
Sum of independent variables	$G_{X+Y}(t) = G_X(t) \times G_Y(t)$
Uniqueness	Matching PGFs guarantee matching distributions

Common exam mistakes

- Forgetting to substitute $t = 1$ (not $t = 0$) when finding $G_X'(1)$ and $G_X''(1)$.
- Using $\text{Var}(X) = G_X''(1) - [G_X'(1)]^2$ without the extra $+ G_X'(1)$ term — this is the single most common PGF error, since it's easy to confuse with the $E(X^2) - [E(X)]^2$ formula from Chapter 1 (remember $G_X''(1)$ gives $E[X(X-1)]$, not $E(X^2)$, directly).
- Adding PGFs instead of multiplying them for a sum of independent variables.
- Failing to simplify $e^{a(t-1)} \times e^{b(t-1)}$ correctly using laws of indices.
- Not checking $G_X(1) = 1$ as a basic sanity check on a derived PGF.

You should now be able to

- Define and derive the probability generating function of a given discrete distribution.
- Quote and derive the PGFs of the Binomial, Poisson and Geometric distributions.
- Use differentiation of a PGF to find $E(X)$ and $\text{Var}(X)$.
- Use the product rule for PGFs to find the distribution of a sum of independent random variables, and justify the result using the uniqueness property.